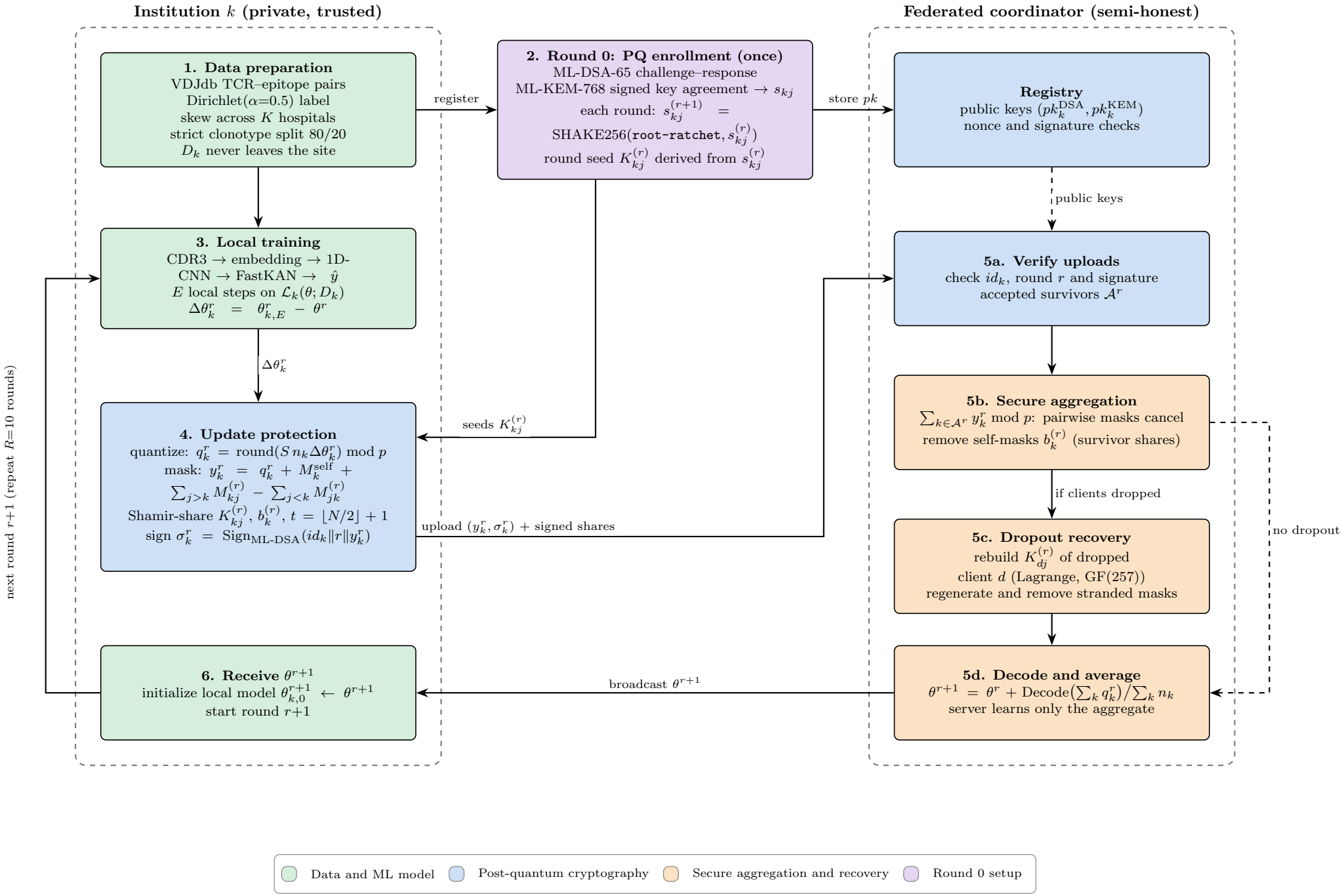

Institution k (private, trusted)
Federated coordinator (semi-honest)
1. Data preparation
VDJdb TCR–epitope pairs
Dirichlet(α=0.5) label
skew across K hospitals
strict clonotype split 80/20
D_k never leaves the site
register
2. Round 0: PQ enrollment (once)
ML-DSA-65 challenge–response
ML-KEM-768 signed key agreement → s_{kj}
each round: $s_{kj}^{(r+1)}$ =
SHAKE256(root-ratchet, $s_{kj}^{(r)}$)
round seed $K_{kj}^{(r)}$ derived from $s_{kj}^{(r)}$
store pk
Registry
public keys ($pk_k^{\mathrm{DSA}}, pk_k^{\mathrm{KEM}}$)
nonce and signature checks
public keys
3. Local training
CDR3 → embedding → 1D-
CNN → FastKAN → $\hat{y}$
E local steps on $\mathcal{L}_k(\theta; D_k)$
$\Delta\theta_k^r = \theta_{k,E}^r - \theta^r$
$\Delta\theta_k^r$
5a. Verify uploads
check id_k, round r and signature
accepted survivors $\mathcal{A}^r$
seeds $K_{kj}^{(r)}$
4. Update protection
quantize: $q_k^r = \mathrm{round}(S\, n_k \Delta\theta_k^r) \bmod p$
mask: $y_k^r = q_k^r + M_k^{\mathrm{self}} +$
$\sum_{j>k} M_{kj}^{(r)} - \sum_{j<k} M_{jk}^{(r)}$
Shamir-share $K_{kj}^{(r)}$, $b_k^{(r)}$, $t = \lfloor N/2 \rfloor + 1$
sign $\sigma_k^r = \mathrm{Sign}_{\mathrm{ML\text{-}DSA}}(id_k \| r \| y_k^r)$
5b. Secure aggregation
$\sum_{k\in\mathcal{A}^r} y_k^r \bmod p$: pairwise masks cancel
remove self-masks $b_k^{(r)}$ (survivor shares)
if clients dropped
upload (y_k^r, σ_k^r) + signed shares
5c. Dropout recovery
rebuild $K_{dj}^{(r)}$ of dropped
client d (Lagrange, GF(257))
regenerate and remove stranded masks
no dropout
next round r+1 (repeat R=10 rounds)
6. Receive θ^{r+1}
initialize local model $\theta_{k,0}^{r+1} \leftarrow \theta^{r+1}$
start round r+1
broadcast θ^{r+1}
5d. Decode and average
$\theta^{r+1} = \theta^r + \mathrm{Decode}(\sum_k q_k^r)/\sum_k n_k$
server learns only the aggregate
Data and ML model
Post-quantum cryptography
Secure aggregation and recovery
Round 0 setup